

3. (a) (i) Friction  $f$  between the tyres and the road.

1A

$$f = \frac{mv^2}{r}$$

1M

$$8000 = \frac{1200 v^2}{45 \text{ m}}$$

$$v = 17.3 \text{ m s}^{-1}$$

1A 3

- (ii) Smaller

1A

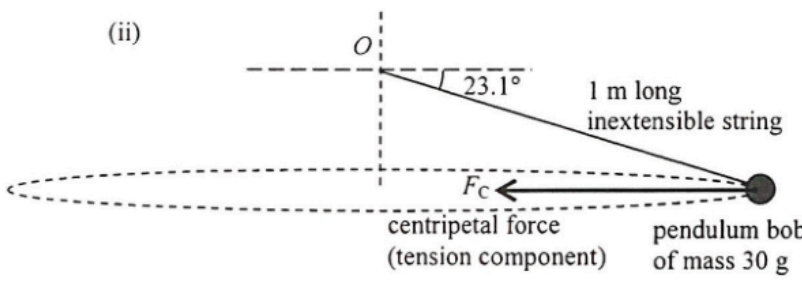
For the same  $f$ ,  $v^2 \propto r$ ; when  $r$  decreases  $v$  decreases.

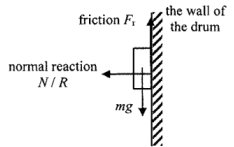
1A 2

- (b) (Max) friction / coefficient of friction reduced,  
not enough to provide the centripetal force / acceleration required for circular motion.  
Or Tracking or allowed speed lowered.

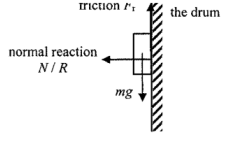
1A

1A 2

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------|
| <p>4. (a) (i) Rotation rate = <math>\frac{\omega}{2\pi} = \frac{5.0}{2\pi}</math><br/> <math>= 0.795775 \text{ (rev s}^{-1}\text{)} \approx 0.80 \text{ (rev s}^{-1}\text{)}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                       | 1M/1A                              | Accept: (0.79 ~ 0.80) rev s <sup>-1</sup> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                                  |                                           |
| <p>(ii)</p>  <p><math>F_C</math> correctly indicated.</p> <p><math>F_C = mr\omega^2</math><br/> <math>= (0.03)(1 \times \cos 23.1^\circ)(5.0)^2</math><br/> <math>= 0.689866 \text{ N} \approx 0.690 \text{ N}</math><br/> <math>(F_C = 0.7033402 \text{ N} \approx 0.703 \text{ N for } g = 10 \text{ m s}^{-2})</math></p> <p>OR</p> <p><math>T \cos \theta = F_C</math> and <math>T \sin \theta = mg</math></p> <p><math>F_C = \frac{mg}{\sin \theta} \cos \theta = 0.689866 \text{ N} \approx 0.690 \text{ N}</math></p> | 1A<br><br>1M<br>1A<br><br>1M<br>1A |                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                                  |                                           |
| <p>(iii) Horizontal component of tension provides the centripetal force, thus tension is larger than the centripetal force.</p> <p>OR <math>T \cos \theta = F_C</math><br/> <math>\Rightarrow T &gt; F_C</math></p> <p>OR <math>T \sin \theta = mg</math><br/> <math>\Rightarrow T (0.750 \text{ N}) &gt; F_C (0.690 \text{ N})</math></p>                                                                                                                                                                                                                                                                    | 1M<br>1A<br><br>1M<br>1A           | 2                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                  |                                           |
| <p>(b) (i) The gravitational force is always perpendicular to the moon's motion / displacement / velocity, thus no work is done on the moon by this force (k.e. unchanged)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                | 1A<br>1A                           |                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                  |                                           |
| <p>(ii) (The claim is incorrect) as, by Newton's third law of motion, gravitational force of the same magnitude (but in opposite direction) is acting on the Earth by the moon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                           | 1A<br>1A                           | Accept: action and reaction pair          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                  |                                           |

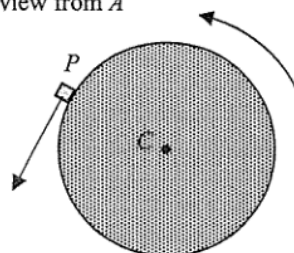
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    |   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
| 4. (a) (i)                                                                 |                                                                                                                                                                                                                                              | 2A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (ii) Normal reaction (force acting on drum/wall of the drum by the object) |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 1 |
| (b) (i)                                                                    | $\omega = 78.5 \text{ rad s}^{-1}$<br>rotation speed = $\frac{78.5}{2\pi} \times 60$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$<br>Or<br>$T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and<br>$f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (ii) centripetal force                                                     | $= m\omega^2 r$<br>$= (2 \times 10^{-5})(78.5)^2(0.25)$<br>$= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$                                                                                                                                                                                                                  | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (iii) $v = r\omega$                                                        | $= 0.25 \times 78.5$<br>$= 19.625 \approx 19.6 \text{ m s}^{-1}$                                                                                                                                                                                                                                                              | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (iv) 1                                                                     |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 1 |

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|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
|                                                                            |                                                                                                                                                                                                                                            | 2A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (ii) Normal reaction (force acting on drum/wall of the drum by the object) |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 1 |
| (b) (i)                                                                    | $\omega = 78.5 \text{ rad s}^{-1}$<br>rotation speed = $\frac{78.5}{2\pi} \times 60$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$<br>Or<br>$T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and<br>$f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (ii) centripetal force                                                     | $= m\omega^2 r$<br>$= (2 \times 10^{-5})(78.5)^2(0.25)$<br>$= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$                                                                                                                                                                                                                  | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (iii) $v = r\omega$                                                        | $= 0.25 \times 78.5$<br>$= 19.625 \approx 19.6 \text{ m s}^{-1}$                                                                                                                                                                                                                                                              | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 2 |
| (iv) 1                                                                     |                                                                                                                                                                                                                                                                                                                               | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                                                                               |    | 1 |

4. (a) (i)  $F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$   
 $= 0.36 \text{ N}$

(ii) Top view from A



(b) (i) same angular speed

(ii) different / smaller magnitude of centripetal acceleration  
 $(a_Q < a_P)$   
 as Q takes a different / smaller radius ( $r_Q < r_P$ )

|    |
|----|
| 1M |
| 1A |
| 2  |
| 1A |
| 1  |
| 1A |
| 1  |
| 1A |
| 1A |
| 2  |